Gaussian Bayesian networks applied to an analysis of environmental pollution and health data in Mexico

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Abstract

This study analyzes health and environmental pollution data in Mexico using Gaussian Bayesian networks (GBNs). The aim is to find and measure the probabilistic relationships between air pollutants and biological health indicators obtained from the ENSANUT survey of 2022. To create a cohesive model, sociodemographic and biomarker data are combined with information from SEMARNAT air quality records.The methodology employed includes variable selection, expert-informed DAG structure creation, and model evaluation using AIC and BIC. Initial results point to significant correlations between exposure to elevated $PM_{2.5}$ and higher CPR values, as well as higher levels of insulin and triglycerides. The method demonstrates how GBNs can help us better understand intricate relationships between pollutants and human health.

1 Introduction

Climate change and air pollution are known to pose critical challenges when it comes to public health. In particular, exposition to gases such as nitrogen dioxide, ozone, and fine particulate matter have been linked to diminished biological biomarkers which are associated with cardiovascular risk, inflammation, and respiratory function. In Mexico, environmental pollution remains a significant health concern, and several measures have been taken in an attempt to mitigate the associated risks as much as possible. However, some of these dangers are not fully understood, and neither are the ways to counteract them.

In particular, exposure to air pollutants such as SO_2, CO, NOx, VOCs, PM_{10}, $PM_{2.5}$, and NH_3 has a direct impact on various biomarkers related to respiratory, cardiovascular, and metabolic health. The main effects include:

- **Particulate matter (PM_{10} and $PM_{2.5}$):** Penetrate the lungs and bloodstream, contributing to systemic inflammation. Key biomarkers affected include C-reactive protein, ferritin , homocysteine, and lipid profiles.
- **Carbon monoxide (CO):** Reduces oxygen transport, leading to hypoxia and affecting hemoglobin and creatinine levels, as well as inflammatory markers.
- **Nitrogen oxides (NOx):** Promote vascular inflammation and accelerate atherosclerosis.Biomarkers impacted include C-reactive protein,LDL cholesterol, total cholesterol, glucose, and insulin levels.
- **Sulfur dioxide (SO_2):** Irritates the respiratory tract and, when converted to sulfates, increases lung and cardiovascular damage. Associated biomarkers include C-reactive protein, triglycerides, cholesterol, and hemoglobin.
- **Volatile organic compounds (VOCs):** Affect liver and kidney function, altering creatinine and uric acid levels, as well as inflammatory biomarkers.
- **Ammonia (NH_3):** Contributes to respiratory and metabolic disorders such as asthma and kidney stress, impacting creatinine, hemoglobin, and micronutrient biomarkers.

Overall, exposure to these pollutants negatively affects multiple systems in the human body, increasing risks for metabolic, cardiovascular, and respiratory diseases, with potential serious health consequences.

Given that these contaminants interact with each other in intricate ways, complex relationships emerge which, in turn, lend themselves to analysis via Gaussian Bayesian networks. Through them, various answers to important questions can be obtained that shed some light on the role these agents play in our modern world.

2 Methodology

Three datasets were used for this analysis: 1) ENSANUT 2022 blood sample data, 2) ENSANUT 2022 socio-demographic data and 3) SEMARNAT air quality measurements for pollutants. Likewise, it was assumed that the people from the first two studies were also exposed to the air pollutants analyzed in the third study. The data sets were merged to ensure consistency of information, and some variables were discarded because of lack of information or because they were irrelevant to the purposes of this analysis. Hence, the final Gaussian Bayesian Networks only used the variables suggested by the medical experts who were consulted. A Gaussian Bayesian network represents dependencies among variables using a directed acyclic graph (DAG). Each variable is modeled as a Gaussian conditional distribution given its parents. the model evaluation used score-based methods (BIC, AIC). Furthermore, catergorical variables such as sex were added so as to enrich the inner structure of the networks.

Once the final network was selected, we addressed queries of public health relevance suggested by the experts. The final variables turned out to be SO_2, NOx, NH_3, CO, COV, PM_010, PM_2_5, valor_PCR, valor_INSULINA, valor_GLU_SUERO, valor_HB1AC, valor_TRIG, valor_COL_LDL, valor_COL_HDL, valor_CREAT and three expert-informed DAGs were proposed were the following:

DAG 1

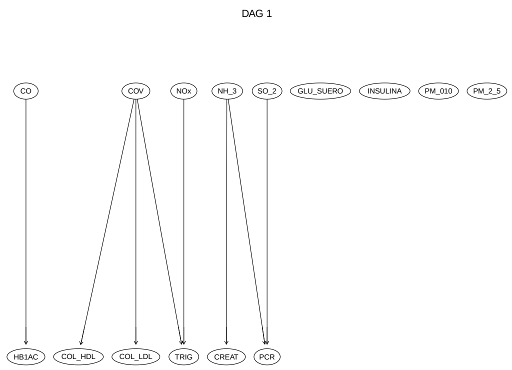

Figure 1: DAG 2

With scores BIC: -169107 and AIC: -169000.6

DAG 2

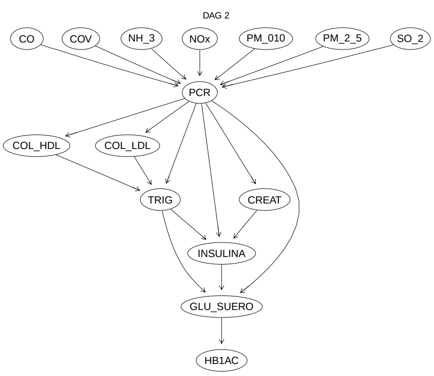

Figure 2: DAG 2

With scores BIC: -168469.1 and AIC: -168329.

DAG 3

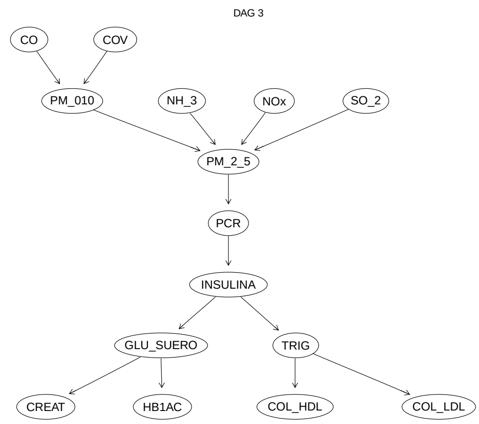

Figure 3: DAG 3

With scores BIC: -165225 and AIC: -165101.8.
As we can see, the DAG that obtained the best results was DAG3

After these results were obtained, an attempt was made to include categorical variables such as sex to the network. However, given the implementation of a Gaussian Bayesian network model, it is not possible to include categorical variables in this model. This is where conditional Gaussian Bayesian networks come in, which adjust the parameters of a linear regression model for the continuous child nodes for each of the categories of their categorical parent nodes. In this type of network, it is assumed that categorical nodes can only be parents of continuous nodes, and continuous nodes cannot be parents of discrete nodes. The final DAG including the sex variable was the following:

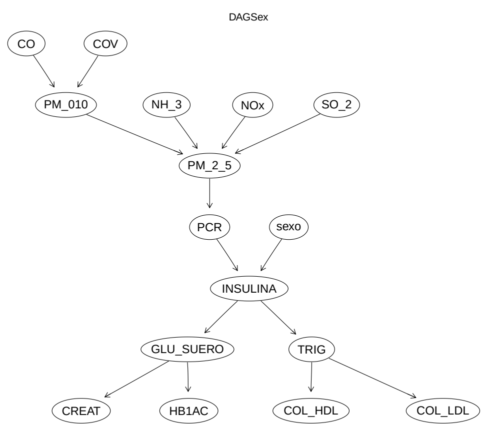

Figure 4: DAG 3 with sex variable

Moreover, a nonparametric model was adjusted, which obtained a final score of -82881.09. As an example, the following graph was created.

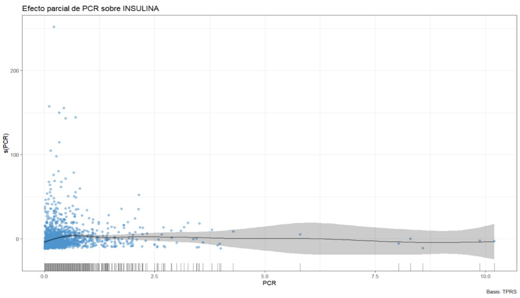

Figure 5: Partial effect insulin and PCR

3 Application

Using the best model, numerical answers were provided to the following key queries:

- What is the probability that a person has a high CRP value if they are exposed to elevated levels of $PM_{2.5}$? Answer: 0.186235

- Given that a person has high insulin levels, what is the probability that they also have elevated triglycerides? Answer 0.4487229

- Between women and men, who has a higher probability of presenting elevated HbA1c values if serum glucose levels are high? Men: 0.6264126, Women: 0.626777

The results indicate that exposure to elevated $PM_{2.5}$ levels is associated with a mild probability of having high CRP values. Additionally, individuals with high insulin levels tend to show a considerably higher probability of also presenting elevated triglycerides. Finally, the comparison between men and women shows almost identical probabilities of having elevated HbA1c when serum glucose levels are high, this indicates that no substantial gender difference can be found in this context.

4 Conclusions

The results revealed that exposure to particulate matter (PM2.5 and PM10), nitrogen oxides (NOx), sulfur dioxide (SO_2), carbon monoxide (CO), volatile organic compounds (VOC) and ammonia (NH_3) is associated with significant changes in biomarkers related to inflammation, cardiovascular risk, and metabolic function, including C-reactive protein, insulin, glucose, hemoglobin A1c, triglycerides and cholesterol fractions. Among the three expert-informed DAGs, DAG 3 demonstrated the best fit according to the BIC and AIC criteria, suggesting that the proposed network structure effectively captures the probabilistic dependencies between pollutants and health indicators.

There were some implications when developing the project, particularly the inclusion of categorical variables such as sex, which required the consideration of conditional Gaussian Bayesian networks. Another notable complication was coordination and communication with medical experts, which sometimes slowed iterative refinement of the network structures and variable selection. Despite these difficulties, the approach proved valuable for quantifying complex interactions in environmental health data and offers a framework for addressing public health questions.

5 References

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